

RozNoor — Hackathon Pitch Deck

Slide-by-slide content plan for the submission presentation

This document lists what should go on each slide of the RozNoor pitch deck, in presentation order. It is a content plan, not the deck itself — use it as the outline when building the actual PowerPoint. Each entry includes the slide's purpose, the exact content/talking points to include, and notes on visuals, screenshots, or diagrams to pair with it.

Recommended total: 14–16 slides for a hackathon demo slot (typically 5–8 minutes of talking time). Keep text on-screen minimal; this plan is deliberately more detailed than what should appear on the slide itself — most of this is spoken narration guidance, not slide text.

SLIDE 1

Title / Cover

Opening slide — sets identity before anyone reads a word.

- Project name: **RozNoor** (include a one-line tagline, e.g. "Silent symptom monitoring for rural Pakistan")
- Team name / hackathon name and date
- Your names (Muhammad Yousif, and the original hackathon teammates Esha, Kashaf, Mishkat — credit the full team even though the build was completed solo)

Visual: App icon (the pulse-concept, teal-palette icon already generated for the Flutter app) large and centered. Clean background, no clutter.

SLIDE 2

The Problem

Establish urgency and stakes before showing any solution.

- Rural/underserved patients often can't or don't report symptoms early — access, literacy, distance to a clinic, or simply not recognizing a symptom as serious
- Chronic disease deterioration (heart failure, diabetes, etc.) is frequently caught too late, when a simple daily check-in could have flagged it days earlier
- Doctors serving these populations have no visibility into a patient's day-to-day status between visits
- State the human cost plainly: delayed care for treatable warning signs

Visual: A simple stat or a short real-world framing sentence in large type. Avoid clutter — one strong statement lands better than a paragraph of statistics here.

SLIDE 3

Our Solution

The single-sentence pitch — this is what judges should remember.

- RozNoor: an AI-powered symptom monitoring and personal baseline health system for rural Pakistani patients
- Patients check in daily by voice or a quick tap-based form — no typing, no literacy barrier
- AI extracts symptoms from natural speech; a disease-specific safety engine classifies risk instantly
- Doctors get a real-time, prioritized dashboard instead of waiting for the next visit

Visual: A simple 4-icon row: Speak → Analyze → Flag → Alert Doctor. Keep it visual, not a wall of text.

SLIDE 4

How It Works — The Core Loop

Walk through the actual product mechanics in one clear diagram.

- Step 1: Voice Diary or Quick Check-in (patient/attendant)
- Step 2: Transcript + AI-powered symptom extraction (Claude Opus 5)

- Step 3: Disease-specific rule-based safety engine (deterministic, not AI-dependent)
- Step 4: Risk result — Green / Yellow / Orange / Red
- Step 5: Doctor dashboard updates instantly; AI-generated summary attached

Visual: This is the single most important diagram in the deck — a horizontal flow diagram with the 5 steps above, icons for each. Consider building this as the Visualizer/diagram artifact separately and exporting it as an image for the slide.

SLIDE 5

Why the Rule Engine Matters (Not Just an AI Wrapper)

Pre-empt the obvious judge question: "is this just a ChatGPT wrapper?"

- The safety-critical risk classification is **pure Python, zero AI dependency** — it works even if the AI key is missing or the API is down
- Hard red-flags (e.g. chest pain) bypass baseline scoring entirely and go straight to Red — no ambiguity, no waiting on a model's confidence score
- AI is used for what it's good at: turning messy natural speech into structured symptom data. The actual safety decision is deterministic and auditable
- This is a deliberate architecture choice — say this explicitly, it's a strong differentiator

Visual: A simple two-box diagram: 'AI Layer (extraction only)' feeding into 'Rule Engine (deterministic, safety-critical)' with an arrow, and a note that the rule engine can run standalone.

SLIDE 6

Personal Baseline Tracking

Show the personalization angle — this isn't a generic threshold checker.

- Three-stage baseline: cold_start → learning → personalized
- Recomputed per-metric on every entry — the system learns what's "normal" for each individual patient
- Example: a +2kg weight gain in 3 days is a heart-failure red flag relative to a patient's own baseline, not a fixed population threshold

Visual: Simple 3-stage progress bar/timeline graphic (cold_start → learning → personalized) with the weight-gain example as a callout.

SLIDE 7

Acoustic Signal Analysis

Your most technically novel feature — give it its own slide.

- Voice entries are analyzed for pause ratio, energy variance, and pitch variability — deterministic signal processing, not machine learning
- Built with numpy + ffmpeg (a real engineering call: librosa was rejected after genuine pip install failures in the deployment sandbox)

- Feeds into the risk result as an **additive-only** signal — it can support a Red flag but can never itself trigger one on its own, preserving the safety-first design

Visual: A simple waveform graphic with the three extracted features labeled, or a short audio-icon-to-data-icon flow.

SLIDE 8 Built for Three Roles, on Two Platforms

Show scope and completeness — this wasn't a single-screen prototype.

- Patient / Attendant — voice diary, quick check-in, results, timeline, weekly trends, profile
- Doctor — patient roster, patient detail with trend charts, alerts, clinical notes
- Admin — user and role management
- All three roles built independently on **both** Flutter (mobile) and React/Vite (web) — same backend, zero duplication of business logic

Visual: A 3x2 grid: three role icons across the top, Flutter/Web logos down the side, or simple screenshots of one screen per role (use the screenshots captured from the live web app here).

SLIDE 9 Screenshots — Patient Experience

Prove it's real and working, not a mockup.

- Show the live Voice Diary / Quick Check-in screen
- Show a real Red-risk result screen (the chest-pain / breathlessness example verified live in production)
- Caption clearly: "Live on production — roznoor.up.railway.app"

Visual: Use the actual screenshots captured from the deployed web app (see the screenshots/ folder). Do not use placeholder mockups here — real, working screenshots are far more persuasive to judges.

SLIDE 10 Screenshots — Doctor & Admin Views

Continue proving completeness across roles.

- Doctor roster/dashboard screen showing the seeded patients and their risk levels
- A doctor viewing the specific alert triggered by the patient screenshot on the previous slide — ties the story together
- Admin people/roles management screen (brief, lower priority than the previous two)

Visual: Screenshots again — same rule, use real captured images from the live deployment, not mockups.

SLIDE 11 Tech Stack & Architecture

Establish technical credibility quickly.

- Mobile: Flutter · Web: React + Vite · Backend: Python FastAPI
- Database: PostgreSQL on Railway (11 tables, zero schema drift across the entire build)
- AI: Direct Anthropic API (Claude Opus 5) — a deliberate quality choice, not a cost default
- Deployment: Railway (backend + web as independent services, both fully live in production)

Visual: A simple architecture diagram: Flutter app and React web app both arrows-into FastAPI backend, backend arrows into Postgres and into Claude API. Keep it clean — 4-5 boxes max.

SLIDE 12 A Deliberate Security Decision

A genuinely good story that shows engineering maturity, not just feature-building.

- A third-party API proxy ("AgentRouter") offered cheap AI credits but its own docs described WAF-evasion and TLS fingerprint spoofing to dodge detection
- Rejected outright for policy and data-integrity reasons — this app handles real health data
- Went with the direct Anthropic API only, confirmed absent from the codebase in a final security sweep

Visual: No visual needed — let this slide be mostly text/quote-styled, it reads as a credibility statement. Optionally a small 'shield' icon.

SLIDE 13 Real Engineering, Real Bugs Fixed

Judges respect honesty about the hard parts — this is not a weakness slide, it's a strength slide.

- Found and fixed a genuine safety gap: voice-only chest pain got zero rule-engine protection when the AI key was unset — confirmed via live phone testing in English and Urdu
- Diagnosed a subtle mic-contention bug (concurrent audio capture from two device APIs) causing 100% silent voice-recognition failures — fixed by sequencing capture instead of running it concurrently
- Root-caused a Postgres connection failure to process-start-order, not the assumed credential mismatch
- (Pick 2-3 of the strongest ones from decisions-log.md — don't try to list all 11, that overwhelms the slide)

Visual: Keep this text-only and confident in tone — this slide's job is to demonstrate real debugging depth, not to apologize for bugs.

SLIDE 14 Live Deployment Status

Prove this isn't a localhost demo — closing technical credibility slide.

- Backend live: roznoor-production.up.railway.app (verified via real seeded-account login returning a valid JWT)
- Web app live: roznoor.up.railway.app (full patient flow verified end-to-end, including a real Red-risk result)

- Mobile release APK built against the same live backend and verified working on a real physical Android device
- State plainly: "Everything you are about to see is running against the real, deployed production system — not a local demo"

Visual: A simple 'status board' style graphic: three rows (Backend / Web / Mobile) each with a green checkmark and their live URL.

SLIDE 15 What's Next

Show self-awareness about scope and a credible next-step story.

- Clinical validation pass on rule-engine thresholds and acoustic-signal cutoffs with a real clinician (currently reasoned-about and internally consistent, but not yet clinically validated)
- Refresh tokens / session revocation (a deliberate MVP scope cut, not an oversight)
- Broader real-device acoustic testing across more network conditions
- (Optional, only if time remains) A lightweight chatbot layer — deliberately out of MVP scope so far

Visual: No visual needed — plain roadmap list, 3-4 items max.

SLIDE 16 Thank You / Questions

Close cleanly, leave contact info and repo visible.

- Thank the judges and hackathon organizers
- Team names again (repeat from slide 1 — some judges only remember the last slide)
- GitHub repo link, and the two live URLs one more time for anyone who wants to try it themselves during Q&A;

Visual: Same app icon from the title slide, for visual bookend consistency.

Presentation Tips

- Rehearse the live-demo portion (slides 9-10's screenshots, or a live click-through if time allows) separately from the slide narration — this is usually where hackathon demos run over time
- If doing a live click-through instead of static screenshots, have the production URLs already open in browser tabs beforehand — don't type them live
- Assign slide 5 (rule engine vs. AI) and slide 12 (security decision) to whoever is most comfortable fielding technical follow-up questions, since these are the most likely to draw judge questions
- Keep slide 13 confident, not apologetic — debugging depth is a selling point in a hackathon context